2ND SEMESTER LAB REPORT

TITLE : TO VERIFY KIRCHHOFF'S VOLTAGE LAW & CURRENTS LAW IN DIFFERENTCONFIGURATION OF DC CIRCUITS



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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 4

Lab Title: VERIFCATION OF OHM'S LAW

Objective: 1TO VERIFY KIRCHHOFF'S VOLTAGE LAW & CURRENTS LAW IN DIFFERENT DC Circuits

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 16/March/2025 Signature:

LABORATORY EXPERIMENT NO. 4

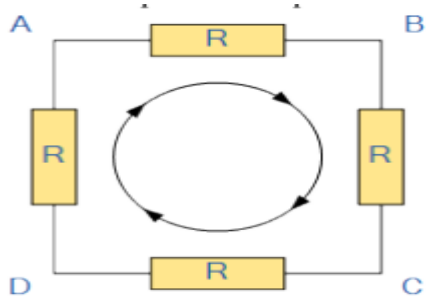
TO VERIFY KIRCHHOFF'S VOLTAGE LAW & CURRENTS LAW IN DIFFERENT CONFIGURATION OF DC CIRCUITS

OBJECTIVE:

1. To verify Kirchhoff's voltage law for dc circuits.
2. To verify Kirchhoff's Current law for dc circuits.

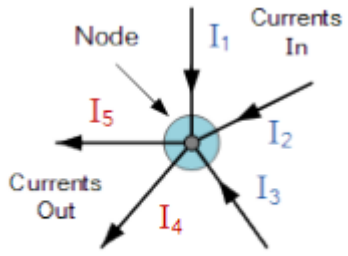
1. Kirchhoffs Voltage Law (KVL)

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$$V_{AB} + V_{BC} + V_{CD} + V_{DA} = 0$$

2. Kirchhoffs Current Law (KCL)

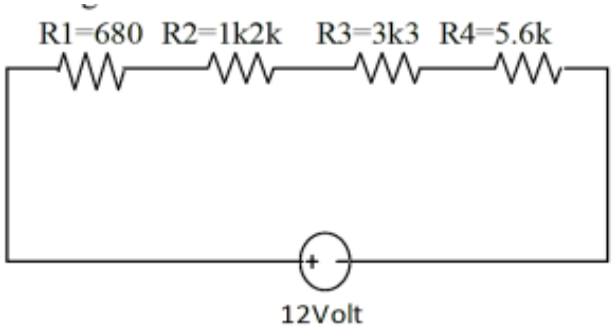


$$I_1 + I_2 + I_3 + (-I_4 + -I_5) = 0$$

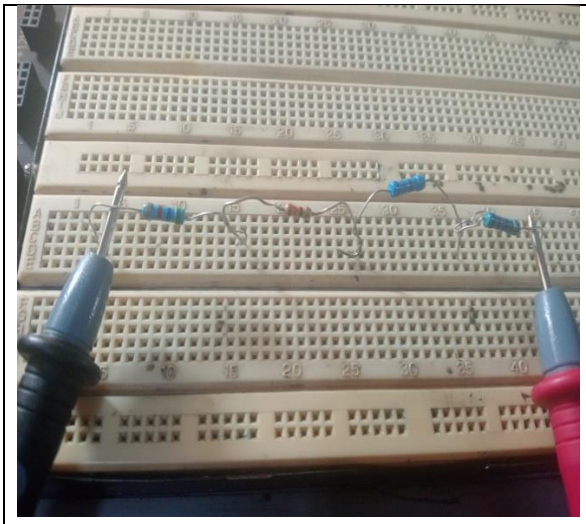
LAB TASKS:

TASK 1. Solve the following circuit by KVL:

(a) Connect the following circuit on breadboard.



R1=6.8K Ω

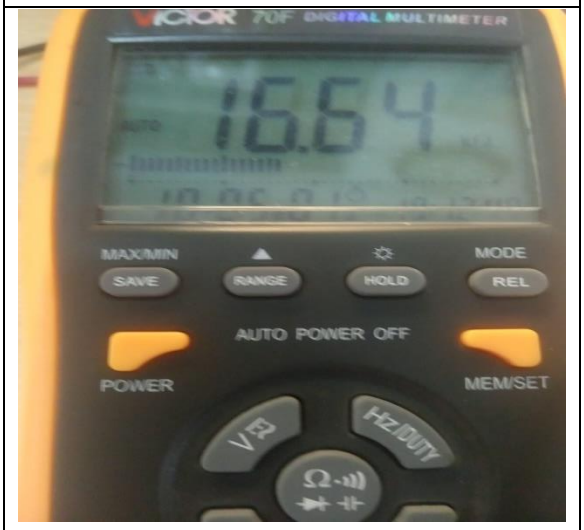


Four resistors of given values are connected in series circuit on breadboard

b) Before connecting the power supply measure the total resistance of the circuit.

Req(measured) =15.9K Ω

Req(calculated) =6.8 kΩ + 1.2 kΩ + 3 kΩ + 5.6 kΩ =6.4K Ω

This is measured equivalent resistance of circuit ion four resistors are connected in series
Measured Resistance


c) Using the measured value of the resistance calculate the current I, voltage across R1, R2, R3 and R4. Record all these values along with the source voltage in the calculated section of the following table. I (calculated) = 0.723 mA

First, let's calculate the total resistance of the circuit:

$R_{total} = R1 + R2 + R3 + R4$

$R_{total} = 6.8\ k\Omega + 1.2\ k\Omega + 3\ k\Omega + 5.6\ k\Omega$

$R_{total} = 6800\ \Omega + 1200\ \Omega + 3000\ \Omega + 5600\ \Omega$

$R_{total} = 16600\ \Omega = 16.6\ k\Omega$

Now, using Ohm's Law, we can calculate the current through the circuit:

$I = V / R_{total}$

$I = 12\ V / 16600\ \Omega$ $I = 0.000723\ A = 0.723\ mA$

So:

The value of current is: 0.723 mA

d) Record all the readings mentioned in the previous steps but this time use a DMM to find the measured values.

e) Compare your results and write an equation verifying KVL.

========== [Theoretical Calculations for KVL (Step-c):]==========

Since this is a series circuit, the same current flows through all resistors. Using Ohm's Law again, we can calculate the voltage across each resistor:

- V(R1) = I × R1 = 0.000723 A × 6800 Ω = 4.92 V
- V(R2) = I × R2 = 0.000723 A × 1200 Ω = 0.87 V
- V(R3) = I × R3 = 0.000723 A × 3000 Ω = 2.17 V
- V(R4) = I × R4 = 0.000723 A × 5600 Ω = 4.05 V

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Table of Results

Circuit Entities	V (Calculated)	V (Measured)
R1	4.92 V	-4.812V
R2	0.87 V	-0.84V
R3	2.17V	2.11V
R4	4.05 V	3.98V
Vs	12V	12V

To verify KVL, the sum of voltage drops around the closed loop should equal the source voltage:

KVL Equation:

the sum of voltage drops around the closed loop should equal the source voltage:

V(R1) + V(R2) + V(R3) + V(R4) = 4.92 V + 0.87 V + 2.17 V = **12.01 V**

This is very close to our source voltage of 12 V (the small difference is due to rounding).

Vs - V(R1) - V(R2) - V(R3) - V(R4) = 0

12V - V(R1) - V(R2) - V(R3) - V(R4) = 0

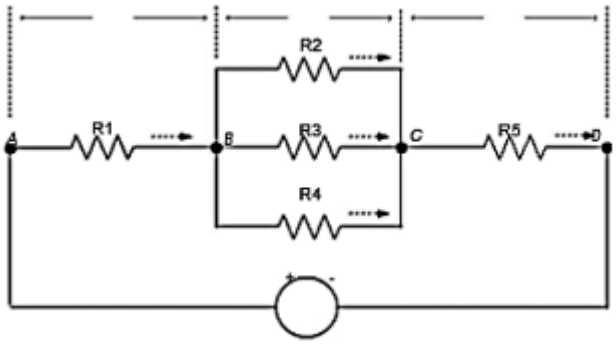
Or rearranged: V(R1) + V(R2) + V(R3) + V(R4) - 12V = 0

With our calculated values: 4.92V + 0.87V + 2.17V + 4.05V - 12V = 12.01V - 12V = 0.01V ≈ 0

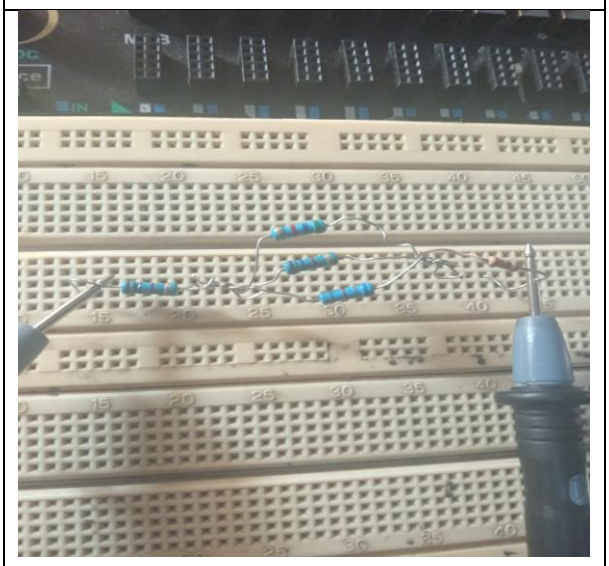
Hence proved KVL.

TASK 2. Solve the following circuit by KCL (at node B):

a) Connect the following circuit on breadboard. Source voltage is 12V and the values of R1, R2, R3, R4 are the same from the previous task. The value of R5 is 1k.



Circuit to prove KCL



b) Before connecting the power supply measure the total resistance of the circuit.

Req(measured) = 8.01K Ω Req(calculated) = 8.96 kΩ

c) Using the measured value of the resistance calculate the source current Is and current through R1, R2, R3, R4 and R5. Record all these values in the calculated section of the following table.

I (calculated) =**1.338 mA**

========== **[Theoretical Calculations for KVL (Step-c):]**==========

Given values:

- R1 = 6.8 kΩ
- R2 = 1.2 kΩ
- R3 = 3 kΩ
- R4 = 5.6 kΩ
- R5 = 1 kΩ
- Vs = 12V

First, I will analyze the equivalent resistance:

1. **R2 and R5 are in parallel:** R2||R5 = (R2×R5)/(R2+R5) = (1.2 kΩ×1 kΩ)/(1.2 kΩ+1 kΩ) = 1200Ω×1000Ω/2200Ω = 545.45Ω
2. **This equivalent resistance (R2||R5) is in series with R3:** R(series) = R2||R5 + R3 = 545.45Ω + 3000Ω = 3545.45Ω
3. **This series combination is in parallel with R4:** R(parallel) =(R(series)×R4)/(R(series)+R4) = (3545.45Ω×5600Ω)/(3545.45Ω+5600Ω) = 2164.54Ω
4. **Finally, this is in series with R1:** Rtotal = R1 + R(parallel) = 6800Ω + 2164.54Ω = 8964.54Ω ≈ 8.96 kΩ

[Final Calculated equivalance resistance is: <u>8.96 kΩ</u>]

Source current: $I_s = V/R_{total}$ = 12V/8964.54Ω = 0.001338 A = 1.338 mA

<u>Source current: = 1.338 mA</u>

- ✓ **R4:** Current through R4: I(R4) = VB/R4 = 2.90V/5600Ω = 0.518 mA
- ✓ **R2/R5:** Combined current through R2 and R5: I(R2+R5) = Is - I(R4) = 1.338 mA - 0.518 mA
- ✓ =0.820 mA
- ✓ **R2:** Current through R2: I(R2) = (VB-VC)/R2 = (2.90V-2.453V)/1200Ω = 0.447V/1200Ω
- ✓ =0.373 mA
- ✓ **R5:** Current through R5: I(R5) = (VB-VC)/R5 = (2.90V-2.453V)/1000Ω = 0.447V/1000Ω
- ✓ =0.447 mA
- ✓ **R3:** Current through R3: I(R3) = I(R2) + I(R5) = 0.373 mA + 0.447 mA = 0.820 mA

Verifying KCL at node B:

I(R1) = I(R2) + I(R4) + I(R5)

1.338 mA = 0.373 mA + 0.518 mA + 0.447 mA 1.338 mA = 1.338 mA

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Table of Results

Circuit Entities	I (Calculated)	I (Measured)
R1	1.338 mA	0.310mA
R2	0.373 mA	0.354mA
R3	0.820 mA	0.810mA
R4	0.518 mA	0.512mA
R5	0.447 mA	0.431mA
Is(Sorce Current)	1.338 mA	1.338mA

KCL equation at node B: I(R1) = I(R2) + I(R4) + I(R5)

Mathematically:

1.338 mA = 0.373 mA + 0.518 mA + 0.447 mA

1.338 mA = 1.338 mA

Hence KCL has proved

Conclusion:

In conclusion, Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL) are fundamental principles in circuit analysis. KVL ensures that the total voltage in a closed loop is conserved, while KCL confirms that the total current at a junction remains balanced. These laws are essential for designing, analyzing, and troubleshooting electrical circuits, including power distribution, electronic devices, and communication systems. By verifying KVL and KCL through practical experiments, we gain a deeper understanding of circuit behavior and enhance our ability to work with electrical systems effectively. Through practical experimentation, we have demonstrated that these laws hold true for DC circuits, validating theoretical calculations with measured values. This verification is crucial for designing and analyzing electrical networks, as KVL and KCL are widely used in circuit design, fault detection, and power system analysis. Their applications extend to electrical engineering, electronics, telecommunications, and embedded systems, making them essential tools for engineers and researchers.